Chapter 16 Intro to Networking Exam

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Score: 100% Passing Score: 80%

Question 1: ✔ Correct

Which Class of Service (COS) priority value should be assigned to a video conference call?

- 3
- 6
- 2
- ➡ 4
- 7
- 0
- 5
- 1

EXPLANATION

A priority value of 4 should be assigned to the video data stream. Each COS priority values specifies a specific traffic type:

- 0 – Background
- 1 – Best effort (default)
- 2 – Excellent effort
- 3 – Critical applications
- 4 – Video (< 100ms latency)
- 5 – Voice (< 10ms latency)
- 6 – Internetwork control
- 7 – Network control

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 2: ✔ Correct

Match each troubleshooting command on the left with its function on the right. Each utility may be used once, more than once, or not at all.

Tests connectivity between two network hosts by sending IPv4 ICMP Echo Request packets without modifying the TTL parameter.

✔ ping

Computes lost/sent packet statistics for each hop in the route between two hosts.

✔ pathping

Used on Linux systems to identify the route between two IPv6 hosts.

✔ traceroute6

Used on Windows systems to identify the route between two IPv4 hosts.

✔ tracert

Tests connectivity between two network hosts by sending IPv6 ICMP Echo Request packets without modifying the TTL parameter.

✔ ping -6

EXPLANATION

Several commonly used network troubleshooting commands include the following:

- The **pathping** command combines the tracert and ping utilities to identify problems at a router or a network link. Unlike tracert or traceroute, pathping can track lost/sent packet statistics for each hop in the route between two hosts. The pathping command is only available on Windows.
- The **ping** command sends an IPv4 ICMP echo request/reply packet to a remote host. A response from the remote host indicates that both hosts are correctly configured and a connection exists between them. The ping command is available on Windows and Linux.
- The **ping -6** command sends an IPv6 ICMP echo request/reply packet to a remote host. A response from the remote host indicates that both hosts are correctly configured and a connection exists between them. The ping -6 command is only available on Windows. On Linux, you would use **ping6** instead.
- The **tracert** command uses ICMP packets to test the path between two IPv4 networks. Responses from each hop on the route are measured three times to provide an accurate representation of how long the packet takes to reach, and be returned by, the destination device. The tracert command is only available on Windows. On Linux, you would use **traceroute** instead.
- The **traceroute6** command is used on Linux systems to identify the route between two IPv6 hosts.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 3: ✔ Correct

You are a network administrator for your company. A user calls and tells you that after stepping on the network cable in her office, that she can no longer access the network.

You go to the office and see that one of the user's stiletto heels has broken and exposed some of the wires in the Cat 5 network cable. You make another cable and attach it from the wall plate to the user's computer.

What should you do next in your troubleshooting strategy?

- ➡ Test the solution.
- Recognize the potential effects of the solution.
- Establish what has changed.
- Document the solution.

EXPLANATION

After you implement a solution, you should always test it. Sometimes more than one problem exists, or your solution may not have solved the problem. In this example, you may have incorrectly set the wires in the network cable you just made, or the user may have jarred the computer and damaged the NIC when she stepped on the cable.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 4: ✔ Correct

Which of the following components do switches use to optimize network performance by performing switching operations in hardware rather than using the CPU and software?

- ➡ An application-specific integrated circuit
- A caching engine
- Ethernet bonding
- A traffic shaper

EXPLANATION

Switches use specialized hardware called an application-specific integrated circuit (ASIC), which performs switching functions in hardware rather than using the CPU and software. ASIC allows switches to perform the switching function at wire speed.

Caching engines are used to store frequently accessed content for faster access; content is retrieved from the local network instead of the internet. Ethernet bonding is used to create two or more physical connections to the same network by bonding NICs or switch ports together; Ethernet bonding provides increased performance and some fault tolerance. A traffic shaper (also called a bandwidth shaper) is a device that is capable of modifying the flow of data through a network in response to network traffic conditions.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 5: ✔ Correct

When troubleshooting network issues, it's important to carry out tasks in a specific order.

Drag the trouble shooting task on the left to the correct step on the right.

Step 1	Step 2	Step 3
✔ Identify the problem.	✔ Establish a theory of probable	✔ Test the theory to determine the cause.
Step 4	**Step 5**	**Step 6**
✔ Establish a plan of action.	✔ Implement the solution or escalate.	✔ Verify full system functionality.
Step 7		
✔ Document findings, actions, and outcomes.		

EXPLANATION

The following is a general approach to network troubleshooting:

1. Identify the problem.
2. Establish a theory of probable cause.
3. Test the theory to determine the cause.
4. Establish a plan of action to resolve the problem and identify potential effects.
5. Implement the solution or escalate as necessary.
6. Verify full system functionality and, if applicable, implement preventative measures.
7. Document findings, actions, and outcomes.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 6: ✔ Correct

A user reports that she can't connect to the Internet. After some investigation, you find that the wireless router has been misconfigured. You are responsible for managing and maintaining the wireless access point.

What should you do next?

- ➡ Create an action plan.
- Determine if escalation is needed.
- Fix the problem.
- Document the problem.

EXPLANATION

At this point, you should create an action plan and account for side effects of the proposed plan. Identifying the affects ahead of time helps you put measures into place to eliminate or reduce any potential negative consequences.

Escalation is not necessary because you are already in charge of managing the wireless access point, and the problem is isolated to that device. Fix the problem only after creating the action plan and identifying possible effects. Document the problem and the solution after the problem has been fixed and the solution has been verified.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 7: ✔ Correct

You are in the process of implementing a network access protection (NAP) infrastructure to increase your network's security.

You are currently configuring the remediation network that non-compliant clients will connect to in order to become compliant. The remediation network needs to be isolated from the secure network.

Which technology should you implement to accomplish this task?

- Virtual private networking (VPN)
- Port security
- Data encryption using PKI
- ➡ Network segmentation

EXPLANATION

Implementing network segmentation would isolate the remediation server from the rest of the network while still allowing the remediation server to contact the NAP infrastructure.

Virtual private networking (VPN) is used to create a secure connection between two hosts or two sites over an unsecured network. Encrypting data transmissions using PKI would only protect transmitted data, not isolate the remediation network. Port security is used to identify allowed and denied devices that connect to a switch port and would not isolate the remediation network.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 8: ✔ Correct

Which of the following devices accepts incoming client requests and distributes those requests to specific servers?

- IPS
- Media converter
- Caching engine
- CSU/DSU
- ➡ Load balancer

EXPLANATION

A load balancer is a device that accepts incoming client requests and distributes those requests to multiple servers. One goal of load balancing is to distribute client requests evenly between multiple servers to improve performance.

A CSU/DSU is a device that converts the signal received from the WAN provider into a signal that can be used by equipment at the customer site. An intrusion prevention system (IPS) can detect and respond to security events. A caching engine saves copies of frequently used content, eliminating the need to download the content each time it is requested. A media converter converts signals used on one media type (such as twisted pair Ethernet) to another media type (such as fiber optic).

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 9: ✔ Correct

Your organization uses a time-keeping application that only runs on Windows 2000 and does not run on newer OS versions. Because of this, there are several Windows 2000 workstations on your network.

Last week you noticed unusual activity on your network coming from the Windows 2000 workstations. After further examination, you discover that the Windows 2000 workstations were the victim of a malicious attack and were being used to infiltrate the network.

You find out that the attackers were able to gain access to the workstations because of the legacy operating system being used. The organization still needs to use the Windows 2000 workstations, which need to be connected to the internet, but you want to make sure the network is protected from future events.

Which solution should you implement to protect the network while also allowing operations to continue as normal?

- Install anti-virus software on the Windows 2000 workstations and configure Windows to automatically download and install updates.
- Create a dedicated network for the Windows 2000 workstations that is completely isolated from the rest of the network, including a separate internet connection.
- Implement a host-based firewall on each Windows 2000 workstation and configure Windows to automatically download and install updates.
- ➡ Configure VLAN membership so that the Windows 2000 workstations are on their own VLAN.

EXPLANATION

The best solution is to place the Windows 2000 workstations in their own VLAN. If you use VLAN network segmentation, the workstations will still have access to the internet, but network access can be heavily restricted. This greatly reduces the damage a workstation can cause if it were to become compromised again.

Legacy operating systems, such as Windows 2000, are easy targets for attackers. This is because legacy operating systems use outdated protocols and have known exploits. Installing an anti-virus or host-based firewall would do very little to protect the entire network. In addition, legacy operating system are no longer supported with updates or patches, so enabling automatic updates would offer no benefit. Creating a dedicated network for the workstations would affect normal operations and also increase network management load.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 10: ✔ Correct

A web server on your network hosts the public website for your company. You want to make sure that a failure of the NIC in the server does not prevent the website from being accessible on the internet.

Which solution should you implement?

- ➡ Ethernet bonding
- Traffic shaping
- QoS
- Spanning tree

EXPLANATION

Ethernet bonding (also called NIC teaming) logically groups two or more physical connections to the same network. If one NIC fails, the second NIC with a connection to the same network can still be used.

Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. A traffic shaper (also called a bandwidth shaper) is a device that is capable of modifying the flow of data through a network in response to network traffic conditions. Quality of Service (QoS) refers to a set of mechanisms that try to guarantee timely delivery or minimal delay of important or time-sensitive communications. QoS is particularly important when implementing Voice over IP (VoIP), Video over IP, or online gaming, where delay or data loss make the overall experience unacceptable.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 11: ✔ Correct

A user is unable to connect to the network. You investigate the problem and determine that the network adapter is defective. You replace the network adapter and verify that it works.

What should you do next?

- Document the problem and solution.
- ➡ Identify the results and effects of the solution.
- Determine if escalation is necessary.
- Create an action plan.

EXPLANATION

After implementing and testing the solution, identify the results and effects of the solution. Make sure that the solution has fully fixed the problem and has not caused any other problems.

Document the problem and solution after it has been fixed and verified. Determine if escalation is needed and create an action plan before implementing the solution.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 12: ✔ Correct

This question includes an image to help you answer the question. View Image

A new assistant network administrator was recently hired by your organization to relieve some of your workload.

You assigned the assistant network administrator to replace a defective patch cable that connected port 1 on your patch panel to one of your network switches. You noticed that it took him an unusually long time to complete this task. Once done, users almost immediately began to report that the network had gone down.

Upon entering the server room, you see that the assistant administrator has configured your network rack as shown in the Exhibit.

What should you do? (Choose two. Each response is a complete solution.)

- [ ] Consolidate all patch cables from the patch panel to a single switch.
- [x] Enable STP on each switch.
- [x] Remove the patch cable connecting the first switch to the third switch.
- [ ] Enable port security on each switch port.
- [ ] Connect the patch panel to the switches with red cross-over cables.
- [ ] Replace the patch cables connecting the switches together with red cross-over cables.

EXPLANATION

The assistant administrator in the scenario appears to have connected the switches together in a way that creates a bridge loop (sometimes called a switching loop). Notice the following:

- Switch1 is connected to Switch2 and Switch3
- Switch2 is connected to Switch1 and Switch3
- Switch3 is connected to Switch1 and Switch2

A bridge loop occurs when there are multiple Layer 2 paths between two network hosts. This usually results in a broadcast storm as the switches repeatedly rebroadcast all broadcast messages, flooding the network.

To fix this issue, you can do one of the following:

- Remove the patch cable connecting the first switch to the third switch. This will break the switching loop and stop the broadcast storm.
- Enable STP on each switch. STP ensures there is only one active path between switches. Switch ports that are part of that path are placed in a forwarding state. Switch ports that are part of redundant but unused paths are placed in a blocking (non-forwarding) state. When an active path goes down, the STP automatically recovers and activates the backup ports necessary to provide continued connectivity.

Consolidating all patch cable from the patch panel to a single switch will not break the bridge loop, nor would enabling port security on each switch port. It is not necessary to replace the patch cables connecting the switches together with cross-over cables, as most switches have Auto-MDIX enabled by default.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 13: ✔ Correct

Which type of switch optimizes network performance by using ASIC to perform switching at wire speed?

- Layer 2 switch
- Layer 1 switch
- ➡ Multilayer switch
- Unmanaged switch

EXPLANATION

A multilayer switch uses specialized hardware called an application-specific integrated circuit (ASIC) to perform switching functions in hardware rather than using the CPU and software. ASIC allows switches to perform the switching function at wire speed.

Layer 2 switches use the CPU and software to forward frames. Unmanaged switches are also called Layer 2 switches. A Layer 1 switch is another name for a hub, which does not perform any traffic inspection; received packets are sent out on all ports.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 14: ✔ Correct

A user reports that she can't connect to a server on your network. You check the problem and find out that all users are having the same problem.

What should you do next?

- Identify the affected areas of the network.
- ➡ Determine what has changed.
- Establish the most probable cause.
- Create an action plan.

EXPLANATION

At this point, you have identified the symptoms and the scope of the problem. In this scenario, you have determined that the problem affects all users. The next step is to determine what might have changed that could have caused the problem.

You have already identified the affected area because you know that the problem affects all users. Before you can choose a probable cause, do additional work to see what might have changed. After selecting a probable cause, determine if escalation is required. Then create an action plan and fix the problem.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 15: ✔ Correct

What is the purpose of using Ethernet bonding? (Select two.)

- [ ] Increases read and write operations between the system bus and network adapters.
- [x] Provides a failover solution for network adapters.
- [x] Increases network performance.
- [ ] Enables Dual Remote Access (DRA) over a WAN link.
- [ ] Provides increased bus speeds.

EXPLANATION

In a true fault tolerant strategy, all system components must be considered. Ethernet bonding (also called adapter teaming) is a fault tolerant strategy that uses multiple network adapters configured in a failover solution. In the event of a NIC failure, other adapters will automatically provide link redundancy. Multiple adapters can also increase performance by distributing the network load between adapters.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 16: ✔ Correct

A user reports that he can't connect to a specific website. You go to the user's computer and reproduce the problem.

What should you do next?

- ➡ Identify the affected areas of the network.
- Determine if escalation is necessary.
- Establish the most probable cause.
- Determine if anything has changed.

EXPLANATION

After identifying the problem, identify the affected area and determine how large the problem is. For example, is the problem limited to this one user, or does it affect all users or a group of users?

After identifying the affected area, find out if anything has changed that might cause the problem. At this point, you should have enough information that you can select a probable cause and determine if escalation is necessary.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 17: ✔ Correct

Your company leases a very fast internet connection and pays for it based on usage. You have been asked by the company president to reduce internet line lease costs. You want to reduce the amount of web pages that are downloaded over the leased connection without decreasing performance.

What is the best way to do this?

- Install a packet-filtering firewall.
- ➡ Install a proxy server.
- Install modems in employees' computers.
- Implement NAT.

EXPLANATION

A proxy server caches frequently visited websites in its cache. It can fulfill client requests from its cache instead of retrieving the information from the internet.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 18: ✔ Correct

A router periodically goes offline. Once it goes offline, you find that a simple reboot puts the router back online.

After doing some research, you find that the most likely cause of the problem is a bug in the router software. A new patch is available from the manufacturer that is supposed to eliminate the problem.

What should you do next?

- Apply the patch to the router.
- Identify the affected areas.
- ➡ Identify possible effects of the solution.
- Document the problem and the solution.

EXPLANATION

Based on the troubleshooting process, you should create an action plan and identify possible effects of implementing the solution. For example, you might consider how long the router will be offline to apply the patch, how long it will take to implement the solution, and what problems that might occur during the process.

Take the action to fix the problem only after you have created the plan, identified the effects, and determined the best time to implement the fix. Document the solution after the problem is fixed and verified. Identifying the affected areas is part of the process you take before arriving at a probable solution.

REFERENCES

LabSim for Network Pro, Section 16.2.

Question 19: ✔ Correct

Drag the broadcast domain property on the left to the appropriate network device(s) on the right. Each property can be used more than once.

Hub

✔ Single broadcast domain

Unmanaged switch

✔ Single broadcast domain

802.11n wireless access point

✔ Single broadcast domain

Router

✔ Multiple broadcast domains

Bridge

✔ Single broadcast domain

Repeater

✔ Single broadcast domain

Layer 3 switch

✔ Multiple broadcast domains

EXPLANATION

A broadcast domain is a logical division of a network. All network hosts within the same broadcast domain can reach each other using broadcasts at the Data Link layer. All network hosts connected to the following Layer 2 network devices are members of the same broadcast domain:

- Hubs
- Unmanaged switches (because they do not support VLANs)
- 802.11 wireless access points
- Bridges
- Repeaters

Layer 3 devices are used to define boundaries between broadcast domains, such as a router or a layer 3 switch. A managed switch with VLANs implemented also creates separate broadcast domain for each VLAN.

REFERENCES

LabSim for Network Pro, Section 16.1.

Question 20: ✔ Correct

Match the class of service (COS) priority on the left with its corresponding value on the right.

0
✔ Background

1
✔ Best effort

2
✔ Excellent effort

3
✔ Critical applications

4
✔ Video (< 100ms latency)

5
✔ Voice (< 10ms latency)

6
✔ Internetwork control

7
✔ Network control

EXPLANATION

Class of service (COS) marks individual frames with a priority value between 0 and 7:

- 0 – Background
- 1 – Best effort
- 2 – Excellent effort
- 3 – Critical applications
- 4 – Video (< 100ms latency)
- 5 – Voice (< 10ms latency)
- 6 – Internetwork control
- 7 – Network control

REFERENCES

LabSim for Network Pro, Section 16.1.

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